

[Assignment VIII]

Problem 2: Consider the optimization problem

$$\underset{\theta \in \Theta}{\text{maximize}} \quad f(\theta).$$

Informally assume f is an intractable function, i.e., evaluating $f(\theta)$ is difficult. However, assume there exists a decomposition

$$f(\theta) = g(\theta, \phi) + h(\theta, \phi) \quad \forall \phi \in \Phi,$$

where g and h are tractable, i.e., evaluating $g(\theta, \phi)$ and $h(\theta, \phi)$ are easy, $h(\theta, \phi) \geq 0$ for all $\theta \in \Theta$ and $\phi \in \Phi$, and for any $\theta \in \Theta$ there exists a $\phi \in \Phi$ such that $h(\theta, \phi) = 0$, i.e., for any $\theta \in \Theta$,

$$\min_{\phi \in \Phi} h(\theta, \phi) = 0$$

and the minimum is attained. Now we consider the following problem with the tractable objective function.

$$\underset{\theta \in \Theta, \phi \in \Phi}{\text{maximize}} \quad g(\theta, \phi).$$

Show that the two optimization problems are equivalent in the sense that

$$\operatorname{argmax} f = \{\theta \mid (\theta, \phi) \in \operatorname{argmax} g\}$$

Hint. Use the fact that

$$\sup_{\theta, \phi} g(\theta, \phi) = \sup_{\theta} \left(\sup_{\phi} g(\theta, \phi) \right).$$

$$\begin{aligned} \tilde{\theta} \in \operatorname{argmax} f &\Rightarrow \exists \tilde{\phi} \in \Phi \quad h(\tilde{\theta}, \tilde{\phi}) = 0, \quad g(\tilde{\theta}, \tilde{\phi}) = f(\tilde{\theta}) - h(\tilde{\theta}, \tilde{\phi}) \leq f(\tilde{\theta}) \leq f(\tilde{\theta}) = g(\tilde{\theta}, \tilde{\phi}) + h(\tilde{\theta}, \tilde{\phi}) = g(\tilde{\theta}, \tilde{\phi}) \\ &\Rightarrow (\tilde{\theta}, \tilde{\phi}) \in \operatorname{argmax} g \end{aligned}$$

$$\begin{aligned} (\tilde{\theta}, \tilde{\phi}) \in \operatorname{argmax} g &\Rightarrow \exists \hat{\phi} \in \Phi \quad h(\tilde{\theta}, \hat{\phi}) = 0, \quad \underbrace{g(\tilde{\theta}, \tilde{\phi})}_{\operatorname{argmax}} = f(\tilde{\theta}) - h(\tilde{\theta}, \tilde{\phi}) = g(\tilde{\theta}, \hat{\phi}) + h(\tilde{\theta}, \hat{\phi}) - h(\tilde{\theta}, \tilde{\phi}) \leq g(\tilde{\theta}, \hat{\phi}) \\ &\Rightarrow h(\tilde{\theta}, \tilde{\phi}) = 0, \quad f(\tilde{\theta}) = g(\tilde{\theta}, \tilde{\phi}) \\ &\Rightarrow \forall \theta \exists \phi \in \Phi \quad h(\theta, \phi) = 0, \quad f(\theta) = g(\theta, \phi) + h(\theta, \phi) = g(\theta, \phi) \leq g(\tilde{\theta}, \tilde{\phi}) = f(\tilde{\theta}) \\ &\Rightarrow \tilde{\theta} \in \operatorname{argmax} f \end{aligned}$$

Therefore those two optimisation problems are equivalent

Problem 3: Inverse permutation. Let S_n denote the group of length n permutations. Note that the map $i \mapsto \sigma(i)$ is a bijection. Define $\sigma^{-1} \in S_n$ as the permutation representing the inverse of this map, i.e, $\sigma^{-1}(\sigma(i)) = i$ for $i = 1, \dots, n$. Describe an algorithm for computing σ^{-1} given σ .

Clarification. In this class, defined σ as a list of length n containing $\{1, \dots, n\}$ exactly once. The output of the algorithm, σ^{-1} , should also be provided as a list.

$$\sigma : [\sigma(1), \sigma(2), \dots, \sigma(n)]^t$$

$$\sigma^{-1} : [(\sigma^{-1}(1), \sigma^{-1}(2), \dots, \sigma^{-1}(n))]^t$$

$$\downarrow [1, \sigma^{-1}(\sigma(1)), 2, \sigma^{-1}(\sigma(2)), \dots, n, \sigma^{-1}(\sigma(n))]^t //$$

$$\downarrow [\sigma(1), \dots, \sigma(n)]^t \text{ or } [1, 2, \dots, n]^t \text{ or } \sigma^{-1}$$

Problem 4: Permutation matrix. Given a permutation $\sigma \in S_n$, the *permutation matrix* of σ is defined as

$$P_\sigma = \begin{bmatrix} e_{\sigma(1)}^T \\ e_{\sigma(2)}^T \\ \vdots \\ e_{\sigma(n)}^T \end{bmatrix} \in \mathbb{R}^{n \times n},$$

where $e_1, \dots, e_n \in \mathbb{R}^n$ are the standard unit vectors. Show

(a) $P_\sigma^T = P_\sigma^{-1} = P_{\sigma^{-1}}$ and

(b) $|\det P_\sigma| = 1$.

Hint. If the rows of $U \in \mathbb{R}^{n \times n}$ are orthonormal, we say U is an orthogonal matrix. Orthogonal matrices satisfy $UU^T = U^TU = I$.

(a) $P_\sigma^T = [e_{\sigma(1)} \dots e_{\sigma(n)}] \Rightarrow [P_\sigma^T P_\sigma]_{ij} = \langle e_{\sigma(i)}, e_{\sigma(j)} \rangle = \delta_{ij}$, $P_\sigma^T P_\sigma = I_n$, $P_\sigma^T = P_\sigma^{-1}$

$P_\sigma^{-1} P_\sigma = I$ $\Leftrightarrow P_\sigma^T P_\sigma^{-T} = I = [e_{\sigma(1)} \dots e_{\sigma(n)}] [e_{\sigma^{-1}(1)} \dots e_{\sigma^{-1}(n)}] ?$

(L) $[e_{\sigma(1)} \dots e_{\sigma(n)}] [e_{\sigma^{-1}(1)} \dots e_{\sigma^{-1}(n)}] e_i$

$= [e_{\sigma(1)} \dots e_{\sigma(n)}] e_{\sigma^{-1}(i)} = e_{\sigma\sigma^{-1}(i)} = e_i \quad \forall i \in \{1, 2, \dots, n\}$

$\Rightarrow$ Linear Extension Theorem of $[e_{\sigma(1)} \dots e_{\sigma(n)}] [e_{\sigma^{-1}(1)} \dots e_{\sigma^{-1}(n)}] = \text{id} = I_n$

$\Rightarrow P_\sigma^{-1} = P_\sigma^{-T} = P_\sigma^T$

(b) $P_\sigma^T P_\sigma = I \Rightarrow \det(P_\sigma^T P_\sigma) = \det(P_\sigma)^2 = \det(I_n) = 1 \Rightarrow |\det P_\sigma| = 1$

(In fact, $\det P_\sigma = \text{sgn } \sigma$. It can be proved using transposition and elementary matrix theory, because every permutation can be represented by compositions of transpositions.)

Problem 5: Affine coupling layer with permutations. Consider the affine coupling layer defined as follows. Let $\Omega \subseteq \{1, \dots, n\}$ and $0 < |\Omega| < n$. Define $\Omega^c = \{1, \dots, n\} \setminus \Omega$. For $x \in \mathbb{R}^n$, define

$$x_\Omega \in \mathbb{R}^{|\Omega|}, \quad x_{\Omega^c} \in \mathbb{R}^{n-|\Omega|}$$

to be the sub-vectors of x with the indices within Ω and Ω^c selected. Define z_Ω and z_{Ω^c} analogously for $z \in \mathbb{R}^n$. The affine coupling layer as

$$z_\Omega = x_\Omega$$

$$z_{\Omega^c} = e^{s_\theta(x_\Omega)} \odot x_{\Omega^c} + t_\theta(x_\Omega),$$

where $s_\theta: \mathbb{R}^{|\Omega|} \rightarrow \mathbb{R}^{n-|\Omega|}$ and $t_\theta: \mathbb{R}^{|\Omega|} \rightarrow \mathbb{R}^{n-|\Omega|}$. Show that

$$\log \left| \frac{\partial z}{\partial x} \right| = \mathbf{1}_{(n-|\Omega|)}^T s_\theta(x_\Omega).$$

Clarification. We are not assuming $|\Omega| = n/2$.

Hint. Find a permutation σ such that

$$\frac{\partial z}{\partial x} = P_{\sigma^{-1}} \begin{bmatrix} I & 0 \\ * & \text{diag}(e^{s_\theta(x_\Omega)}) \end{bmatrix} P_\sigma.$$

Let $\sigma \in S_n$ be an arbitrary permutation such that σ satisfies $\sigma(1), \dots, \sigma(|\Omega|) \in \Omega$

Reorder x, z as x_σ, z_σ .

$$\frac{\partial z}{\partial x} = \frac{\partial z_\sigma}{\partial x_\sigma} \frac{\partial x_\sigma}{\partial x} = P_\sigma \begin{bmatrix} I & 0 \\ * & \text{diag}(e^{s_\theta(x_\Omega)}) \end{bmatrix} P_\sigma$$

($\because \frac{\partial z_i}{\partial x_j} = f_{ij}$ ($1 \leq i, j \leq |\Omega|$) $\rightarrow$ const.)

$$\frac{\partial z_i}{\partial x_j} = \frac{\partial}{\partial x_j} (e^{\text{const.}} x_i \cdot x_i + t_\theta(x_\Omega)_{x_i})$$

$$= f_{ij} e^{s_\theta(x_\Omega)_{x_i}}$$

$$\left| \frac{\partial z}{\partial x} \right| = \cancel{\det P_\sigma^{-1}} \det I \det(\text{diag}(e^{s_\theta(x_\Omega)})) \cancel{\det P_\sigma}$$

$$\log \left(\frac{\partial z}{\partial x} \right) = \sum_{i=1}^{n-|\Omega|} s_\theta(x_\Omega)_{x_i} = \mathbf{1}_{(n-|\Omega|)}^T s_\theta(x_\Omega)$$

(ex) $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix}$

$$\frac{\partial x_\sigma}{\partial x} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix} = P_\sigma$$

Problem 8: Log-derivative trick for VAE. Let $Z \in \mathbb{R}^k$ be a random variable. Let $q_\phi(z)$ be a probability density function for all $\phi \in \mathbb{R}^p$. Assume $q_\phi(z)$ is differentiable in ϕ for all fixed $z \in \mathbb{R}^k$. Let $h: \mathbb{R}^k \rightarrow \mathbb{R}$ satisfying $h(z) > 0$ for all $z \in \mathbb{R}^k$. Assume that the order of integration and differentiation can be swapped. Show

$$\nabla_\phi \mathbb{E}_{Z \sim q_\phi(z)} \left[\log \left(\frac{h(Z)}{q_\phi(Z)} \right) \right] = \mathbb{E}_{Z \sim q_\phi(z)} \left[(\nabla_\phi \log q_\phi(Z)) \log \left(\frac{h(Z)}{q_\phi(Z)} \right) \right].$$

Hint. Since $q_\phi(z)$ is a probability density function,

$$\int \nabla_\phi q_\phi(z) dz = \nabla_\phi \int q_\phi(z) dz = \nabla_\phi 1 = 0.$$

$$\nabla_\phi \mathbb{E}_{Z \sim q_\phi(z)} \left[\log \left(\frac{h(Z)}{q_\phi(Z)} \right) \right] = \nabla_\phi \int \log \frac{h(z)}{q_\phi(z)} q_\phi(z) dz = \nabla_\phi \left(\int \log h(z) q_\phi(z) dz - \int \log q_\phi(z) q_\phi(z) dz \right)$$

$$= \int \log h(z) \nabla_\phi q_\phi(z) dz - \left(\int \frac{\nabla_\phi q_\phi(z)}{q_\phi(z)} q_\phi(z) dz + \int (\nabla_\phi q_\phi(z)) \log q_\phi(z) dz \right)$$

$$= \int \log h(z) \nabla_\phi q_\phi(z) dz - \left(\cancel{\nabla_\phi \int q_\phi(z) dz}^{=1} + \int (\nabla_\phi q_\phi(z)) \log q_\phi(z) dz \right)$$

$$= \int (\nabla_\phi q_\phi(z)) \log \frac{h(z)}{q_\phi(z)} dz = \int \frac{\nabla_\phi q_\phi(z)}{q_\phi(z)} \log \frac{h(z)}{q_\phi(z)} q_\phi(z) dz$$

$$= \int (\nabla_\phi \log q_\phi(z)) \log \frac{h(z)}{q_\phi(z)} q_\phi(z) dz$$

$$= \mathbb{E}_{Z \sim q_\phi(z)} \left[(\nabla_\phi \log q_\phi(Z)) \log \frac{h(Z)}{q_\phi(Z)} \right] \quad \square$$